KAZAKH-BRITISH TECHNICAL UNIVERSITY

KBTU

**JSC «Kazakh-British Technical University»**
**School of IT and Engineering**

**APPROVED BY**
**Dean of SITE**
**Azamat Imanbayev**
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«08» January 2024

**SYLLABUS**

**Discipline:** "IT infrastructure and Computer Networks"
**Number of credits:** 6 (2/0/2)
**Term:** Spring 2024
**Instructor's full name:** Mels Begenov, Azamat Imanbayev, Temirlan Zhaxalykov

| Personal Information about the Instructor | Time and place of classes | | Contact information | |
|---|---|---|---|---|
| | Lessons | Office Hours | Tel.: | e-mail |
| Mels Begenov, Master of Engineering | According to the schedule | After lecture | | m.begenov@kbtu.kz |
| Azamat Imanbayev, Master of Engineering | According to the schedule | | | a.imanbaev@kbtu.kz |
| Temirlan Zhaxalykov, Master of Engineering | According to the schedule | | | t.zhaxalykov@kbtu.kz |

**Course duration:** 6 ECTS, 15 weeks, 60 class hours.

**Course prerequisites:** "ICT"

**Course description:**
This course introduces the architecture, structure, functions, components, and models of the Internet and other computer networks. The principles and structure of IP addressing, and the fundamentals of Ethernet concepts, media, and operations are introduced to provide a foundation for the curriculum. By the end of the course, students will be able to build simple LANs, perform basic configurations for routers and switches, and implement IP addressing schemes.

**Course objectives**
Students who complete "Computer Networks" will be able to perform the following functions:

- Understand and describe the devices and services used to support communications in data networks and the Internet
- Understand and describe the role of protocol layers in data networks
- Understand and describe the importance of addressing and naming schemes at various layers of data networks in IPv4 and IPv6 environments
- Design, calculate, and apply subnet masks and addresses to fulfill given requirements in IPv4 and IPv6 networks
- Explain fundamental Ethernet concepts, such as media, services, and operations
- Build a simple Ethernet network using routers and switches
- Use command-line interface (CLI) commands to perform basic router and switch configurations
- Utilize common network utilities to verify small network operations and analyze data traffic

**Literature**

**Required**

1. Cisco Networking Academy “CCNA Introduction to Networks” curriculum
2. Олифер Виктор, Олифер Наталья. Компьютерные сети. Принципы, технологии, протоколы: Юбилейное издание. — СПб.: Питер, 2021. — 1008 с.:
3. Doug Lowe. Networking All-in-One Desk Reference For Dummies: 8th Edition. Wiley, 2021
4. James Kurose, Keith Ross. Computer Networking: A Top-Down Approach, 7th Edition. Pearson, 2017
5. Mike Meyers. CompTIA Network+ Certification All-in-One Exam Guide. McGraw-Hill Education, 2018
6. Andrew S. Tanenbaum, “Computer Networks”, 4rd Edition, Prentice Hall PTR, 2003.

## COURSE CALENDAR

| **Week** | **Class work** | | | **SIS** | | **TSIS** | |
|---|---|---|---|---|---|---|---|
| | **Topic** | **Lectures** | **Labs** | **Task** | **Hours** | **Task** | **Hours** |
| 1 | **<u>Lecture 1</u>: Networking Today**<br>Networks Affect our Lives<br>Network Representations and Topologies<br>Common Types of Networks<br>Internet Connections<br>Reliable Networks<br>Network Trends<br>Network Security<br>**<u>Packet Tracer 1:</u>** 1.5.7 - Packet Tracer - Network Representation<br>**<u>Lab 1:</u>** Research IT and Networking Job Opportunities | 2 | 2 | | 1 | | 1 |
| 2 | **<u>Lecture 2:</u> Basic Switch and End Device Configuration**<br>Cisco IOS Access<br>IOS Navigation<br>The Command Structure<br>Basic Device Configuration<br>Save Configurations<br>Ports and Addresses<br>Configure IP Addressing<br><br>**<u>Packet Tracer 2:</u>**<br>**2.3.7 - Packet Tracer - Navigate the IOS**<br>**2.5.5 - Packet Tracer - Configure Initial Switch Settings**<br>**2.7.6 - Packet Tracer - Implement Basic Connectivity**<br>**2.9.1 - Packet Tracer - Basic Switch and End Device Configuration**<br>**<u>LAB 1:</u>**<br>2.3.8 - Lab - Navigate the IOS by Using Tera Term for Console Connectivity<br>2.9.2 - Lab - Basic Switch and End Device Configuration | 2 | 2 | | 1 | | 1 |
| 3 | **<u>Lecture 3:</u> Protocols and Models**<br>The Rules<br>Protocols<br>Protocol Suites<br>Standards Organizations<br>Reference Models | 2 | 2 | | 1 | | 1 |

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| | Data Encapsulation<br>Data Access<br>**Packet Tracer 3:**<br>**3.5.5 - Packet Tracer - Investigate the TCP/IP and OSI Models in Action**<br>**LAB 3:**<br>3.0.3 - Class Activity - Design a Communications System<br>3.4.4 - Lab - Research Networking Standards<br>3.7.9 - Lab - Install Wireshark<br>**3.7.10 - Lab - Use Wireshark to View Network Traffic** | | | | | | |
| 4 | **Lecture 4: Physical Layer/ Number Systems**<br>Purpose of the Physical Layer<br>Physical Layer Characteristics<br>Copper Cabling<br>UTP Cabling<br>Fiber-Optic Cabling<br>Wireless Media<br>Binary Number System<br>Hexadecimal Number System<br>**Packet Tracer 4:**<br>**4.6.5 - Packet Tracer - Connect a Wired and Wireless LAN**<br>**4.7.1 - Packet Tracer - Connect the Physical Layer**<br>**LAB 4:**<br>**4.6.6 - Lab - View Wired and Wireless NIC Information** | 2 | 2 | | 1 | | 1 |
| 5 | **Lecture 5: Data Link Layer/Ethernet Switching**<br>Purpose of the Data Link Layer<br>Topologies<br>Data Link Frame<br>Ethernet Frames<br>Ethernet MAC Address<br>Switch Speeds and Forwarding Methods<br>**LAB 5:**<br>**7.1.6 - Lab - Use Wireshark to Examine Ethernet Frames**<br>**7.2.7 - Lab - View Network Device MAC Addresses**<br>**7.3.7 - Lab - View the Switch MAC Address Table** | 2 | 2 | | 1 | | 1 |
| 6 | **Lecture 6: Network Layer**<br>Network Layer Characteristics<br>IPv4 Packet<br>IPv6 Packet<br>Media Access Control (MAC)<br>Introduction to Routing | 2 | 2 | | 1 | | 1 |
| 7 | **Lecture 7: Address Resolution/ Basic Router Configuration**<br>MAC and IP<br>Address Resolution Protocol (ARP)<br>IPv6 Neighbor Discovery<br>Configure Initial Router Settings<br>Configure Interfaces, Default Gateway<br>**Packet Tracer 7:**<br>**9.1.3 - Packet Tracer - Identify MAC and IP** | 2 | 2 | | 1 | | 1 |

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| | **Addresses**<br>**9.2.9 - Packet Tracer - Examine the ARP Table**<br>**9.3.4 - Packet Tracer - IPv6 Neighbor Discovery**<br>**10.1.4 - Packet Tracer - Configure Initial Router Settings**<br>**10.3.4 - Packet Tracer - Connect a Router to a LAN**<br>**10.3.5 - Packet Tracer - Troubleshoot Default Gateway Issues**<br>**10.4.3 - Packet Tracer - Basic Device Configuration**<br>**<u>LAB 7:</u>**<br>**10.4.4 - Lab - Build a Switch and Router Network** | | | | | | |
| 8 | **Midterm** | 2 | 1 | | 1 | | 1 |
| 9 | **<u>Lecture 8:</u> IPv4 Addressing**<br>IPv4 Address Structure<br>IPv4 Unicast, Broadcast, and Multicast<br>Network Segmentation<br>Subnet an IPv4 Network<br>VLSM<br>Structured Design<br>**<u>Packet Tracer 8:</u>**<br>**11.5.5 - Packet Tracer - Subnet an IPv4 Network**<br>**11.7.5 - Packet Tracer - Subnetting Scenario**<br>**11.9.3 - Packet Tracer - VLSM Design and Implementation Practice**<br>**11.10.1 - Packet Tracer - Design and Implement a VLSM Addressing Scheme**<br>**<u>LAB 8:</u>**<br>**11.6.6 - Lab - Calculate IPv4 Subnets**<br>**11.10.2 - Lab - Design and Implement a VLSM Addressing Scheme** | 2 | 2 | | 1 | | 1 |
| 10 | **<u>Lecture 9:</u> IPv6 Addressing/ICMP**<br>IPv4 Issues<br>IPv6 Address Representation<br>IPv6 Address Types<br>GUA and LLA Static Configuration<br>Subnet an IPv6 Network<br>ICMP Messages<br>Ping and Traceroute Tests<br>**<u>Packet Tracer 9:</u>**<br>**12.6.6 - Packet Tracer - Configure IPv6 Addressing**<br>**12.9.1 - Packet Tracer - Implement a Subnetted IPv6 Addressing Scheme**<br>**13.2.6 - Packet Tracer - Verify IPv4 and IPv6 Addressing**<br>**13.2.7 - Packet Tracer - Use Ping and Traceroute to Test Network Connectivity**<br>**13.3.1 - Packet Tracer - Use ICMP to Test and Correct Network Connectivity**<br>**<u>LAB 9:</u>**<br>12.7.4 - Lab - Identify IPv6 Addresses<br>12.9.2 - Lab - Configure IPv6 Addresses on Network Devices<br>13.3.2 - Lab - Use Ping and Traceroute to Test | 2 | 2 | | 1 | | 1 |

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| | Network Connectivity | | | | | | |
| 11 | **Lecture 10: Transport Layer**<br>Transportation of Data<br>TCP/UDP Overview<br>Port Numbers<br>TCP Communication Process<br>Reliability and Flow Control<br>**Packet Tracer 10:**<br>**14.8.1 - Packet Tracer - TCP and UDP Communications** | 2 | 2 | | 1 | | 1 |
| 12 | **Lecture 11: Application Layer**<br>Application, Presentation, and Session<br>Peer-to-Peer<br>Web and Email Protocols<br>IP Addressing Services<br>File Sharing Services<br>**LAB 11:**<br>15.4.8 - Lab - Observe DNS Resolution | 2 | 2 | | 1 | | 1 |
| 13 | **Lecture 12: Network Security Fundamentals**<br>Security Threats and Vulnerabilities<br>Network Attack Mitigations<br>Device Security<br>**Packet Tracer 12**:<br>**16.4.6 - Packet Tracer - Configure Secure Passwords and SSHl**<br>**16.5.1 - Packet Tracer - Secure Network Devices**<br>**LAB 12:**<br>16.2.6 - Lab - Research Network Security Threats<br>16.4.7 - Lab - Configure Network Devices with SSH<br>**16.5.2 - Lab - Secure Network Devices** | 2 | 2 | | 1 | | 1 |
| 14 | **Lecture 13: Building a Small Network**<br>Devices in a Small Network<br>Small Network Applications and Protocols<br>Scale to Larger Networks<br>Verify Connectivity<br>Host and IOS Commands<br>Troubleshooting Methodologies<br>Troubleshooting Scenarios<br>**Packet Tracer 13:**<br>**17.5.9 - Packet Tracer - Interpret show Command Output**<br>**17.7.7 - Packet Tracer - Troubleshoot Connectivity Issues**<br>**17.8.2 - Packet Tracer - Skills Integration Challenge**<br>**17.8.3 - Packet Tracer - Troubleshooting Challenge**s<br>**LAB 13:**<br>17.4.6 - Lab - Test Network Latency with Ping and Traceroute<br>17.7.6 - Lab - Troubleshoot Connectivity Issues<br>17.8.1 - Lab - Design and Build a Small Business Network | 2 | 2 | | 1 | | 1 |
| 15 | **End of term** | 2 | 1 | | 1 | | 1 |
| | **Total** | 30 | 30 | | 15 | | 15 |

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| | **Final Exam** |

**Course assessment parameters**

| Type of activity | 1 | 2 |
|---|---|---|
| Lab works | 6 * 2 = 12 | 6 * 2 = 12 |
| Attendance / participation | 0 | 0 |
| Midterm / end of term | 18 | 18 |
| | **30** | **30** |
| Final exam | 40 | |
| **Total** | **100** | |

**Criteria for evaluation of students during semester**

| No | Assessment criteria | Weeks | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | |
| 1. | Lab works | | * | * | * | * | * | * | | * | * | * | * | * | * | | 24 |
| 2. | Attendance / participation | * | * | * | * | * | * | * | * | * | * | * | * | * | * | * | 0 |
| 3. | Midterm / end of term | | | | | | | | * | | | | | | | * | 36 |
| 4. | Final exam | | | | | | | | | | | | | | | | 40 |
| | Total | | | | | | | | | | | | | | | | 100 |

**ATTENTION!**
If student missed more than 30% of lessons student receives «F (Fail)» grade;
If for two attestations student receives 29 or less points, this student is not accepted to final exam and for all course he (she) receives «F (Fail)» grade;
If student receives on final exam 19 or less points, then independently on how many points he (she) received for two attestations, in whole he (she) receives «F (Fail)» grade;
In the case of missing or being late for final exam without plausible reason, independently on how many points he (she) received for two attestations, in whole he (she) receives «F (Fail)» grade.
It is forbidden to change the topic of the course project and change the composition of the team after 2 weeks of training.
If a student obtains 30 points in theoretical knowledge, but does not have a finished course project (15 week), he is not allowed to defend the project and receives an «F (Fail)» grade for the course automatically.
If a student missed more than 50% of the lectures due to health problems and has medical documents in the form of KBTU, but did not complete the course project, the student is not allowed to defend the course project, and it is recommended to take an academic leave.
In case of non-compliance of the course project with the given assignment, the student is not allowed to defend the course project and automatically receives an «F (Fail)» grade.
In case of detection of plagiarism in the course project, the student is automatically not allowed to defend the course project and receives «F (Fail)» grade.
At the exam, the student must prepare a printed course project, a presentation and a software implementation of the project. If any documents are missing, the student does not automatically give the right to defend the course project and receives an «F (Fail)» grade.
The delivery of the electronic version of the course project and presentations in MS TEAMS should be no later than 15 weeks, if students do not upload the course project on time, they will not automatically finish the exam and receives an «F (Fail)» grade.

**Academic Policy:**
**Cheating, duplication, falsification of data, plagiarism are not permitted under any circumstances!**
Students must participate fully in every class. While attendance is crucial, merely being in class does not constitute “participation”. Participation means reading the assigned materials, coming to class prepared to ask questions and engage in discussion.
Students are expected to take an active role in learning (the instructor will provide the information and guidelines to do this).
Students must come to class on time.
Students are to take responsibility for making up any work missed.
Make up tests in case of absence will not normally be allowed.
Mobile phones must always be switched off in class.
Students should always show tolerance, consideration and mutual support towards other students.

**Students are encouraged to**
consult the teacher on any issues related to the course;
make up within a week’s time for the works undone for a valid reason without any grade deductions;
make any proposals on improvement of the academic process;
track down their continuous rating throughout the semester.

***Master of Engineering*** *Mels Begenov*

***Minutes #34 of School of Information Technology and Engineering meeting on January 8, 2024***